Case Study 1: Attendance Recovery Computation

1. Problem Statement

Develop a Python program to calculate a student's current attendance percentage and determine the minimum number of consecutive classes the student must attend to reach the required attendance percentage of 75%. The program should also display the new attendance percentage after attending the required number of classes.

2. Objectives

- To calculate the current attendance percentage.
- To check whether a student meets the required attendance percentage.
- To determine the minimum number of additional classes needed to reach 75% attendance.
- To display the updated attendance percentage.
- To understand the use of variables, operators, conditional statements, and loops in Python.

3. Assumptions

1. Total classes conducted so far are 60.
2. Classes attended by the student are 40.
3. The required attendance percentage is 75%.
4. The student attends every upcoming class.
5. Every additional class increases both the total classes and the attended classes by one.
6. The input values are valid, and the total number of classes is greater than zero.

4. Algorithm

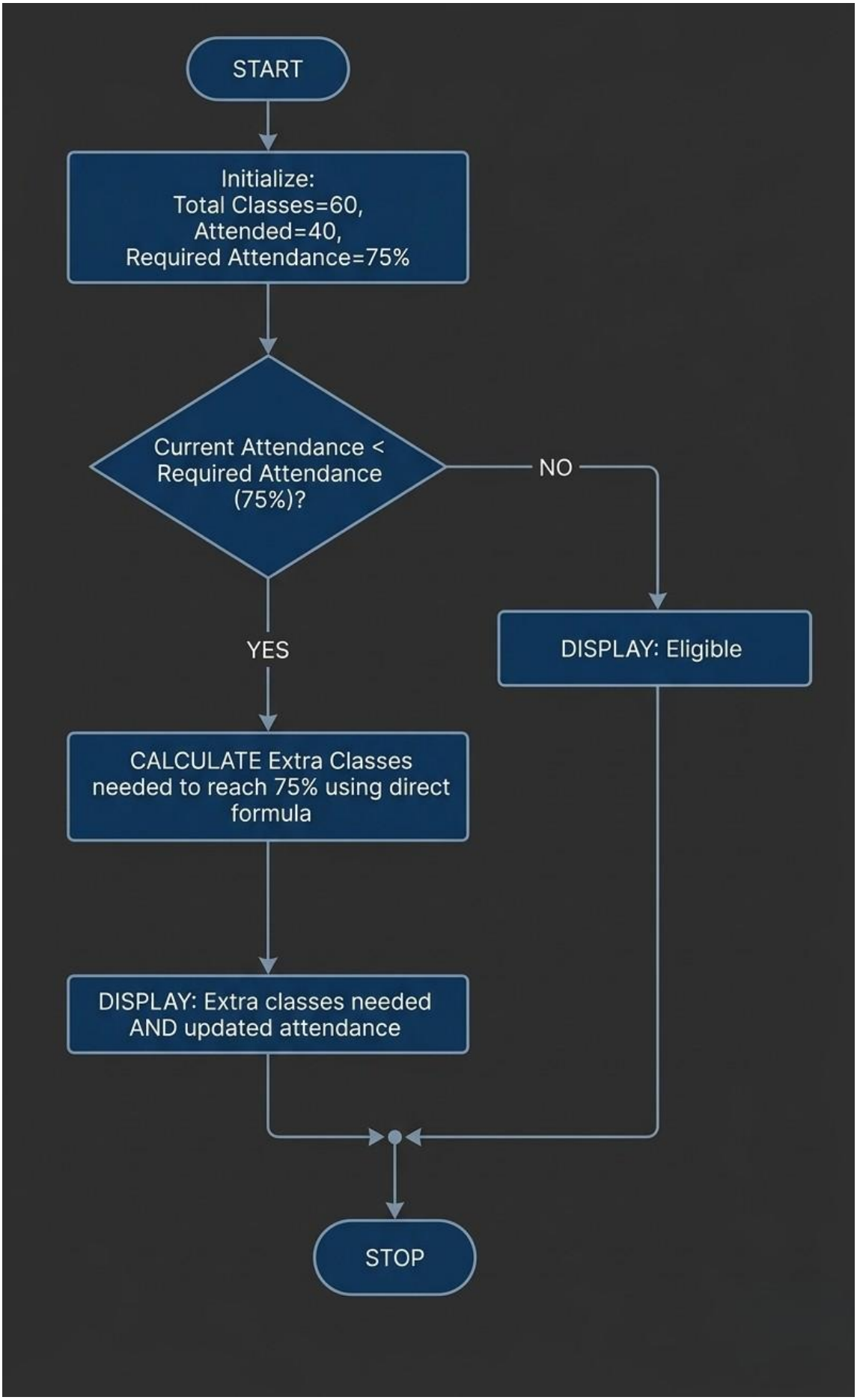
1. Start the program.

2. Initialize the total number of classes, attended classes, and required attendance percentage.
3. Calculate the current attendance percentage using the formula:

   Attendance= Classes attended/Total classes x 100

4. Display the current attendance percentage.
5. Check whether the attendance percentage is at least 75%.
6. If yes, display that the student is eligible and no additional classes are required.
7. Otherwise, initialize the number of additional classes to zero.
8. Use a while loop to increase the number of additional classes until the required attendance percentage is reached.
9. Calculate the updated attendance percentage.
10. Display the number of additional classes required and the updated attendance percentage.
11. Stop the program.

5.Flowchart


START
Initialize:
Total Classes=60,
Attended=40,
Required Attendance=75%
Current Attendance <
Required Attendance
(75%)?
NO
YES
DISPLAY: Eligible
CALCULATE Extra Classes
needed to reach 75% using direct
formula
DISPLAY: Extra classes needed
AND updated attendance
STOP

6. Pseudocode

```
START

INPUT total_classes
INPUT attended_classes

IF total_classes <= 0 OR attended_classes < 0
  OR attended_classes > total_classes THEN
   DISPLAY "Invalid input"
ELSE
   SET required_percentage = 75

   CALCULATE attendance =
      (attended_classes / total_classes) * 100

   DISPLAY current attendance

   IF attendance >= required_percentage THEN
      DISPLAY "Eligible"
      DISPLAY "Additional classes required = 0"
   ELSE
      SET extra_classes = 0

      WHILE ((attended_classes + extra_classes) /
```

```
        (total_classes + extra_classes)) * 100
        < required_percentage DO
      SET extra_classes = extra_classes + 1
    END WHILE

    CALCULATE new_attendance =
      ((attended_classes + extra_classes) /
      (total_classes + extra_classes)) * 100

    DISPLAY extra_classes
    DISPLAY new_attendance
  END IF
END IF

STOP
```

7. Source code

attendance recovery.py

```python
total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))
required_percentage = 75

if total_classes <= 0 or attended_classes < 0 or attended_classes > total_classes:
    print("Invalid input! Please enter valid class details.")

else:
    attendance = (attended_classes / total_classes) * 100

    print("Current attendance:", round(attendance, 2), "%")

    if attendance >= required_percentage:
        print("Attendance Requirement: Eligible")
        print("Additional classes required: 0")

    else:
        extra_classes = 0

        while ((attended_classes + extra_classes) /
               (total_classes + extra_classes)) * 100 < required_percentage:
            extra_classes += 1

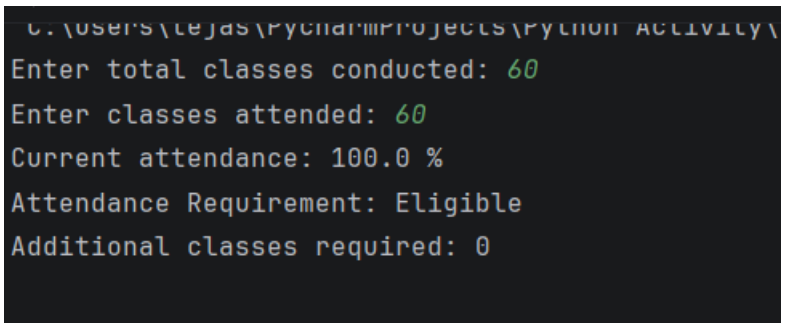
        new_attendance = (
            (attended_classes + extra_classes) /
            (total_classes + extra_classes)
        ) * 100

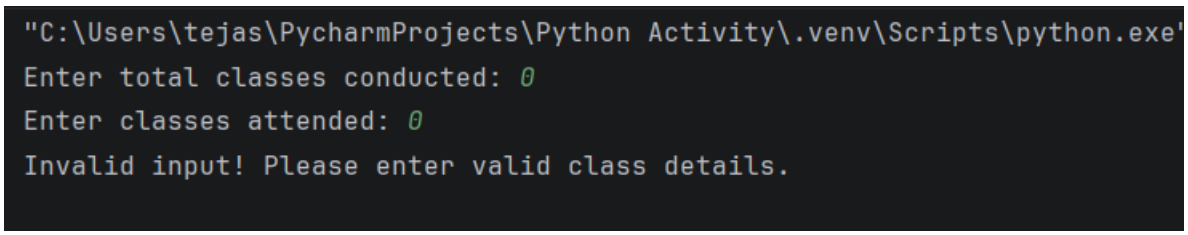
        print("Attendance Requirement: Not Eligible")
        print("Additional classes to attend:", extra_classes)
        print("Updated attendance:", round(new_attendance, 2), "%")
```

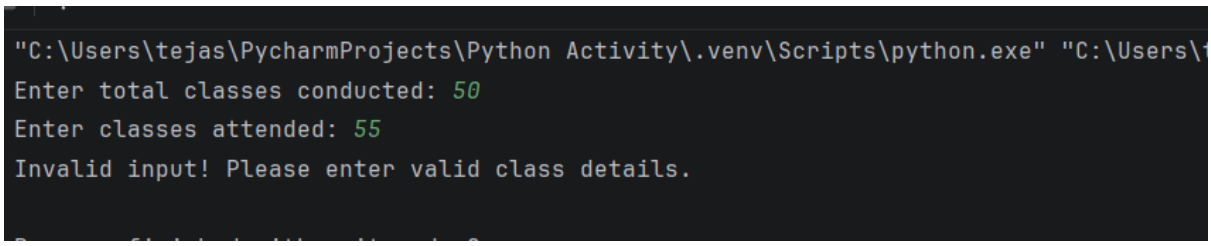
8. Test cases

Total classes	Classes attended	Expected result
60	40	20 additional classes
60	60	Already eligible
0	0	Invalid input
50	55	Invalid input

9. Output







10. Conclusion

The Attendance Recovery Computation program calculates the current attendance percentage and determines the minimum number of additional classes required to reach the specified attendance threshold. It uses arithmetic operators, conditional statements, and a while loop to solve a practical academic problem.

This case study helps demonstrate the basic Python concepts included in the prescribed syllabus and shows how programming can be used to solve real-life problems.